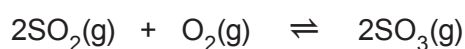


5. (a) Sulfuric acid is manufactured using the Contact Process. The equation represents one stage of the process.



Describe and explain **all** the stages in the process, including the one given above, from **raw materials** to the final product, **sulfuric acid**. [6 QER]

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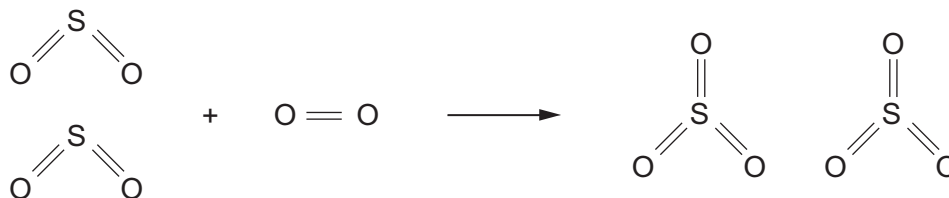
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- (b) The equation shows the bonds which are broken and the bonds which are formed in the production of sulfur trioxide.



The bond energy of an $\text{S}=\text{O}$ bond is 523 kJ.

- (i) The **total** energy needed to break the bonds in the reactants is 2587 kJ. Calculate the energy needed to break an $\text{O}=\text{O}$ bond. [2]

Energy = kJ

- (ii) Calculate the overall energy change for the reaction. [2]

Overall energy change = kJ

- (iii) Complete the energy profile for the reaction. [1]

